

TrES-5b

R-0835

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_180418 · created 2026-07-28T18:04:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458226.9117 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.024
Transit depth (Rp/Rs)^2	1.48 +/- 0.59 [%]
Orbital Inclination (inc)	83.5 +/- 2.09
Airmass coefficient 1 (a1)	1.085 +/- 0.015
Airmass coefficient 2 (a2)	-0.0475 +/- 0.0056
Scatter in the residuals of the lightcurve fit is	1.4 %
Best Comparison Star	#1 - [145, 75]
Optimal Method	PSF photometry
Transit Duration (day)	0.063 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.121 ± 0.024 vs 0.142 (deviation 0.81σ)
Duration fit vs published	0.063 d vs 0.0758 d (deviation 0.66σ)
Mid-time O–C	-1.0 min
Depth SNR	4.7
Residual scatter	1.4 %

Light curve

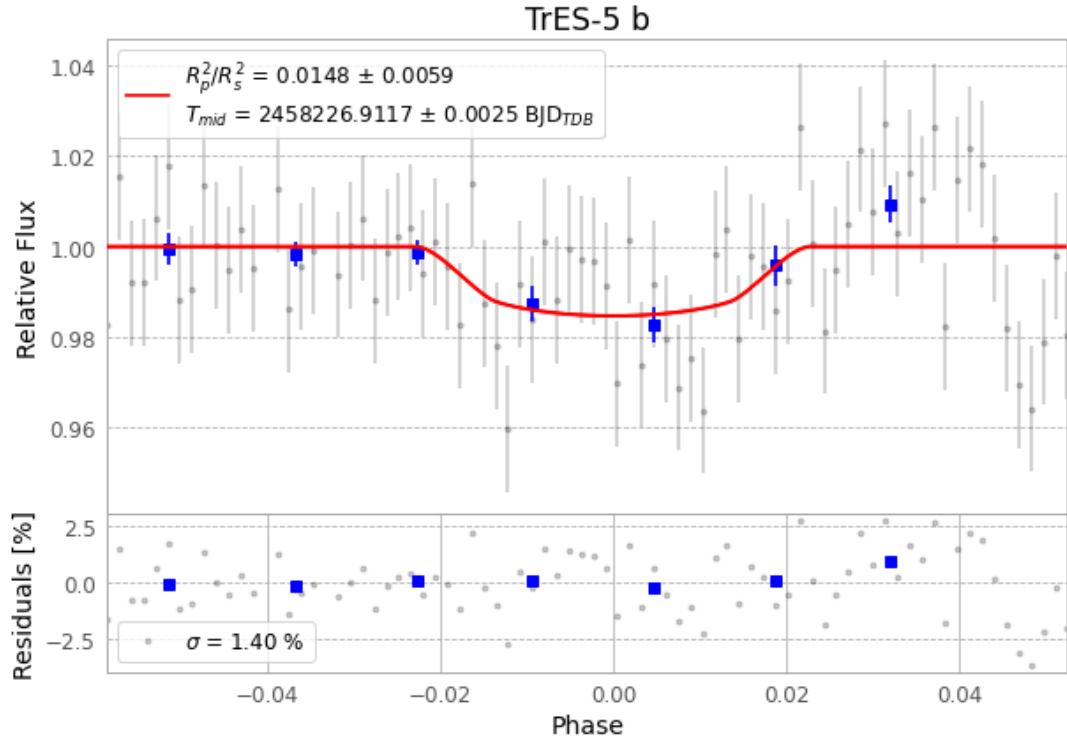


Figure 1 — FinalLightCurve_TrES-5 b_2018-04-18.png (SHA-256 232f8e451e8ca634...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [441, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 29e3b0dc8d70ae53...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

80 FITS frames (53 MB), TRES-5180418074812.FITS → TRES-5180418114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-180418.log (11ded7d77e72...), FinalLightCurve_TrES-5 b_2018-04-18.png (232f8e451e8c...), FinalLightCurve_TrES-5 b_2018-04-18.csv (8fe8b1d22334...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 18:04Z.